4 A technician is designing a charging circuit for a compact camera flash unit as shown in Fig. 4.1. The circuit consists of a DC battery V of 12 V, a resistor R = 15 kΩ and an uncharged capacitor C of 220 μF, connected in series with a switch.

At time t = 0, the switch is closed.

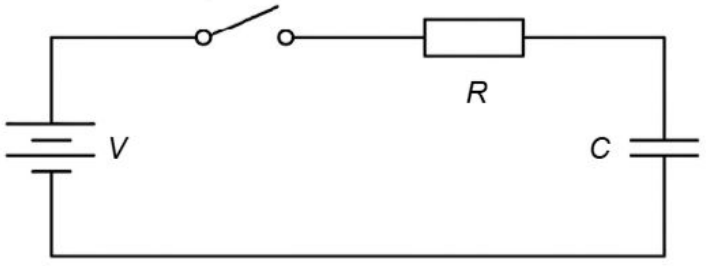


Fig. 4.1

(a) Calculate the time constant for the circuit.

The time constant of a series RC circuit is the product of the resistance and the capacitance:

$RC = 15 \times 10^3 \times 220 \times 10^{-6} = 3.3$ s [A1]

time constant = ... s [1]

(b) Calculate the percentage of maximum charge in the capacitor 6.6 s after the switch is closed.

$$Q = Q_0\left(1 - e^{-\frac{t}{RC}}\right) \Rightarrow \frac{Q}{Q_0} = \left(1 - e^{-\frac{6.6}{3.3}}\right)$$ [B1]

$$= 0.8647 \approx 87\%$$ [A1]

percentage = ... % [2]

(c) Hence or otherwise, determine the current flowing through the circuit exactly 6.6 s after the switch is closed.

$$Q \propto V_C \Rightarrow \frac{V_C}{V_0} = 0.8647$$ [B1]

$V_C = 0.8647(12) = 10.376$ V

$V_R = 12 - V_C = 1.624$ V

$$I = \frac{V_R}{R} = \frac{1.624 \text{ V}}{15 \times 10^3\ \Omega} = 0.108 \text{ mA}$$ [A1]

alternatively:

charging current, $I = I_0 e^{-\frac{t}{RC}}$

$$I = \frac{V_0}{R}\left(e^{-\frac{6.6}{3.3}}\right) = 0.108 \text{ mA}$$

current = ... mA [2]

[Total: 5]